

LABLAB.AI × ALPACA — AI TRADING AGENTS HACKATHON

An options desk run by *agents*.

A momentum-scanning, options-trading pipeline that watches 168 tickers, decides through a chain of LLM and deterministic agents, and executes on Alpaca — talking to the market only through Alpaca's own MCP server.

Paper account **PA39Q320P4PP**

Started **\$100,000**

● Live now

Three rules, non-negotiable.

The hackathon's requirements shaped the architecture from day one — none of them were optional, so the pipeline was built to satisfy all three at the protocol level, not just in spirit.

AUTONOMY

The agent has to act on Alpaca's Trading API on its own — no human clicking "submit" per trade.

MCP, NOT REST

Every Alpaca call — quotes, chains, orders, positions — goes through Alpaca's official MCP server, the same tool-calling surface an LLM uses.

OPTIONS, NOT EQUITY

The strategy has to trade options contracts, with real strike/expiry/greeks decisions — not just buy-and-hold shares.

Six stages, one chain per cycle.

Only two stages actually need a model's judgment. The other four are exact math or a fixed tool-call sequence — so that's what they are, in plain code.



DETERMINISTIC – 4 STAGES

Is the market open. Rank 168 symbols by momentum, keep the top 3. Pick a contract by delta, size the position. Submit the order, manage the exit. Zero judgment calls — so zero model latency, zero model failure mode.

LLM-DRIVEN – 2 STAGES

Sentiment reads the news. Strategy picks the trade shape. Both run on Groq, with model fallback and a second account for overflow.

Judgment, clamped in code.

Both LLM agents are free to reason, but neither can act outside a boundary that's enforced by Python — not just asked for in the prompt.

SENTIMENT

Reads recent news for a candidate. Can only push the **NO_TRADE** — it can veto, but it can never invent a trade or `_clamp_signal` . momentum signal toward flip direction. Enforced by

STRATEGY

Given the signal + sentiment + momentum strength, picks a long call, a long put, a risk-reversal combo, or no trade. Can `_clamp_strategy` . never point a different direction than the signal it was handed. Enforced by

INFRA

Groq-hosted, with automatic fallback across several models per call, then a second, genuinely separate Groq account once the first is exhausted for a turn.

Six checks between a signal and a fill.

POSITION SIZE

1% of equity per trade, capped at 5 contracts.

CYCLE CAP

5% of equity max new risk across every candidate in one cycle.

LIQUIDITY

Reject any contract with no bid, or spread over 25% of mid.

DELTA TARGETING

Strike picked by proximity to 0.30 delta, not a flat % — adapts to each name's own volatility.

STOP-LOSS HOLD

30-minute minimum hold before a stop-loss can fire, so spread noise can't trigger it.

COMBO SEQUENCING

A 2-leg combo's short leg only fires after the long leg is confirmed filled.

The bugs that only a live account finds.

Each of these only surfaced against real Alpaca and Groq responses — not something a mock would have caught.

Take-profit / stop-loss asymmetry

+10% take-profit against a -20% stop-loss meant every losing streak outweighed every win. Fixed to a symmetric $\pm 20\%$.

Option chain exhausted by one side

`get_option_chain`'s result limit is shared across the whole strike range — calls alone could fill it before a single put, or a genuinely OTM strike, was ever reached. Fixed by fetching each option type in its own one-sided range.

Groq's token cap is org-wide, not per key

A second API key on the *same* Groq account shares one budget. Real relief needed a genuinely separate account — confirmed live against an actual 429 on Groq's daily token cap.

No server, no shortcuts.

30 / 10 min

Full pipeline every 30 minutes, exit checks every 10 — both GitHub Actions jobs, fired by an external scheduler, capped at a 10-minute timeout so a hang can't eat a slot silently.

131 tests

Fully mocked — strategy math, the LLM tool-calling loop, every agent's direction/veto clamp, and the full orchestrator wiring — no live credentials needed to verify correctness.

SECRETS

Alpaca and Groq credentials live only in GitHub Actions secrets — `.env` is gitignored and checked before every push.

DEPLOY

Docker + docker-compose for local runs; GitHub Actions for the scheduled live pipeline — no server to provision or babysit.

Running right now, on a clean account.

Not a backtest, not a demo mode — a dedicated paper account created after the hackathon's kickoff, trading on schedule with no manual intervention.

FIELD	VALUE
Account	PA39Q320P4PP
Opened at	\$100,000.00
Current equity	\$99,334.38
Universe scanned / cycle	168 symbols
Candidates carried forward / cycle	top 3

● Both scheduled workflows completing on time, every cycle, as of this write-up.

Everything's open.

Public repo, live run history, and the one-page write-up — no login required.

Source — github.com/khalilgrassa-cell/ALPACA-AI-TRADING-HACKATHON



Live runs — scheduled workflow history



One-page write-up — [SUBMISSION.md](#)

